



Laboratorio No.1 – Semana 14 - Integración servicios en la nube

Nombre: _Kenneth Isaías Yahid López Castillo_____ **Carné:** _1138025_____

Objetivo: Desarrollar y publicar una aplicación web que permita interactuar con archivos almacenados en Azure Blob Storage, utilizando control de versiones con Git y una canalización CI/CD de Azure DevOps para automatizar el despliegue de la aplicación en Azure App Service

Resultado esperado

- Aplicación web funcional ejecutándose en Azure App Service.
- Conexión entre App Service y Azure Blob Storage.
- Listado y consulta de archivos almacenados en un contenedor Blob.
- Operación de carga de archivos al Blob Storage.
- Repositorio Git con historial de cambios.
- Pipeline de Azure DevOps que compile y despliegue automáticamente la aplicación.
- Evidencias que demuestren el funcionamiento de la solución.

Repositorio recomendado

Utilice como punto de partida el ejemplo oficial de Microsoft/Azure Samples revisado para esta actividad:

GitHub: <https://github.com/Azure-Samples/serviceconnector-webapp-storageblob-dotnet>

El tutorial oficial utiliza una aplicación web .NET conectada a Azure Blob Storage mediante una identidad administrada y Service Connector. La documentación indica además que la aplicación puede desplegarse en Azure App Service.

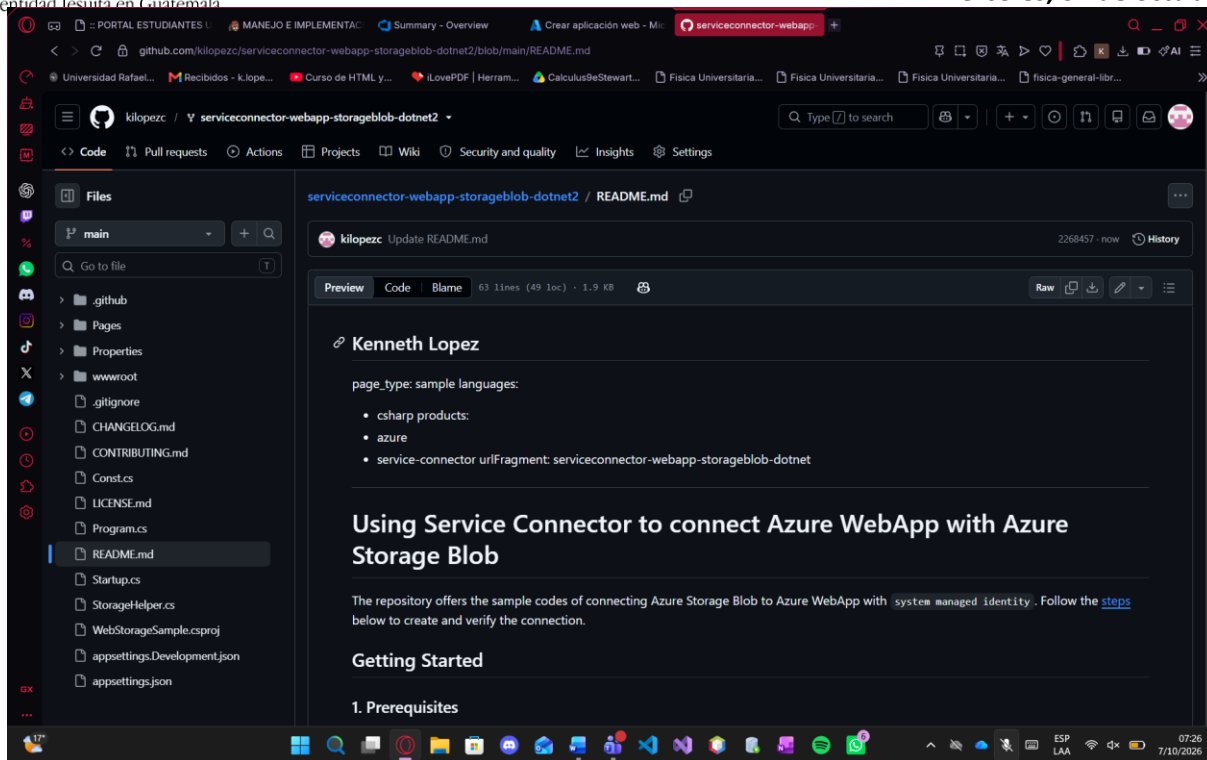
Tecnologías

Tecnología	Uso
C# / .NET	Desarrollo de la aplicación web.
Azure Blob Storage	Almacenamiento de archivos.
Azure App Service	Hospedaje de la aplicación.
Git	Control de versiones.
Azure DevOps Pipelines	Integración y despliegue continuo.
Managed Identity / Service Connector	Acceso seguro desde App Service al almacenamiento.

Realización de la actividad

1. Preparar el repositorio

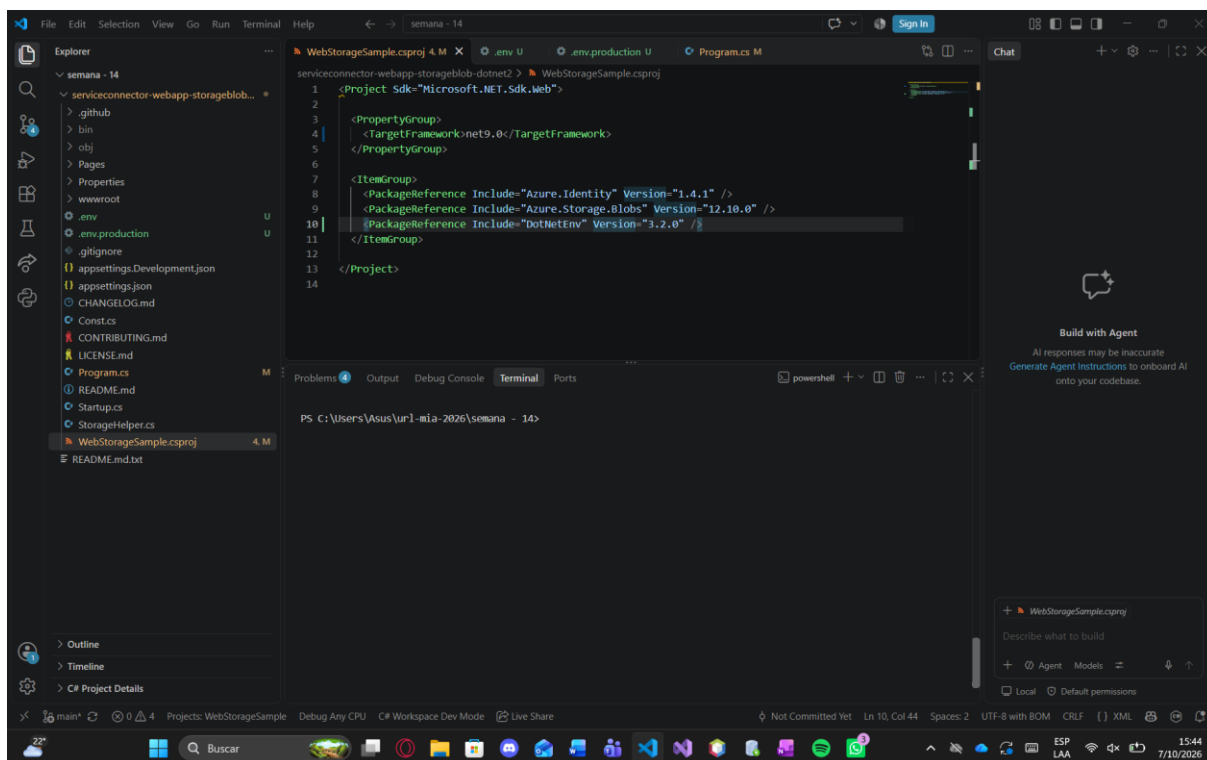
En el repositorio brindado arriba, realice un fork del repositorio base o copie el proyecto a un repositorio propio. El repositorio debe pertenecer al equipo de trabajo. Realice un primer commit identificable



2. Revisar la aplicación:

Ejecute la aplicación localmente y revise cómo obtiene acceso al Blob Storage. Identifique el punto del código donde se realiza la lectura/escritura de blobs y documente brevemente su funcionamiento. Visual Studio -> carpeta -> MIA_Semana14 -> git clone [repo url] -> dotnet commands

NOTA: cambia ren .csproj file -> <TargetFramework>net9.0</TargetFramework>





3. Crear DotNet Env

a) En terminal, instalar: **dotnet add package DotNetEnv**

```
PS C:\Users\ASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2> dotnet add package DotNetEnv

Compilación realizado correctamente en 0.5s
info : La validación de la cadena de certificados X.509 utilizará el almacén de confianza predeterminado seleccionado por .NET para la firma de código.
info : La validación de la cadena de certificados X.509 utilizará el almacén de confianza predeterminado seleccionado por .NET para la marca de tiempo.
info : Agregando PackageReference para el paquete "DotNetEnv" al proyecto "C:\Users\ASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2\WebStorageSample.csproj".
info : GET https://api.nuget.org/v3/registration5-gz-semver2/dotnetenv/index.json
info : OK https://api.nuget.org/v3/registration5-gz-semver2/dotnetenv/index.json 474 ms
info : Restaurando paquetes para C:\Users\ASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2\WebStorageSample.csproj...
info : CACHE https://api.nuget.org/v3/vulnerabilities/index.json
info : CACHE https://api.nuget.org/v3-vulnerabilities/2026.10.06.23.47.38/vulnerability.base.json
info : CACHE https://api.nuget.org/v3-vulnerabilities/2026.10.06.23.47.38/2026.10.07.11.47.40/vulnerability.update.json

C:\Users\ASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2>
```

```
C:\Users\ASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2> dotnet add package DotNetEnv

C:\Users\ASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2> dotnet build
Microsoft (R) Build Engine version 17.0.0 for .NET
Copyright (c) Microsoft Corporation. All rights reserved.

Build Succeeded
    WebStorageSample.csproj : warning NU1902: El paquete "Azure.Identity" 1.4.1 tiene una vulnerabilidad de gravedad moderada conocida, https://github.com/advisories/GHSA-m5v6-6rth-3y9f
    WebStorageSample.csproj : warning NU1902: El paquete "Azure.Identity" 1.4.1 tiene una vulnerabilidad de gravedad moderada conocida, https://github.com/advisories/GHSA-m5v6-6rth-3y9f
    WebStorageSample.csproj : warning NU1902: El paquete "Azure.Storage.Blobs" 12.10.0 tiene una vulnerabilidad de gravedad moderada conocida, https://github.com/advisories/GHSA-64x4-9hc6-r2h6
    WebStorageSample.csproj : warning CA2200: Si inicia de nuevo la excepción detectada, se cambia la información de la pila. (https://learn.microsoft.com/dotnet/fundamentals/code-analysis/quality-rules/ca2200)
    WebStorageSample.csproj : warning CA2200: Si inicia de nuevo la excepción detectada, se cambia la información de la pila. (https://learn.microsoft.com/dotnet/fundamentals/code-analysis/quality-rules/ca2200)

Compilación correcto con 10 advertencias en 1.9s

Hay actualizaciones de carga de trabajo disponibles. Ejecute "dotnet workload list" para obtener más información.
PS C:\Users\ASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2>
```



b) Crear .env file en raíz de proyecto, valores ->

AZURE_STORAGEBLOB_RESOURCEENDPOINT=https://miarchivos2026.blob.core.windows.net/

AZURE_STORAGE_CONTAINER_NAME=default

The screenshot shows the Visual Studio Code interface. The Explorer pane on the left shows the project structure with files like .env, .env.production, and various .cshtml files. The main editor shows the .env file with the following content:

```
1 AZURE_STORAGEBLOB_RESOURCEENDPOINT=https://miarchivos2026.blob.core.windows.net/  
2 AZURE_STORAGE_CONTAINER_NAME=default  
3
```

The Terminal pane at the bottom shows the output of the dotnet build command, indicating a successful compilation with 10 warnings and 1 error. The warnings are related to the Azure.Identity and Azure.Storage.Blobs packages.

c) Crear .env.production con los mismos valores en el mismo lugar

d) Añadir cambio brindado en program.cs

The screenshot shows the Visual Studio Code interface with the Program.cs file open in the main editor. The file contains the following code:

```
1 using Microsoft.AspNetCore.Hosting;  
2 using Microsoft.Extensions.Hosting;  
3 using Microsoft.Extensions.Hosting;  
4  
5 namespace WebStorageSample  
6 {  
7     0 references  
8     public class Program  
9     {  
10         0 references  
11         public static void Main(string[] args)  
12         {  
13             Env.Load(".env");  
14             CreateHostBuilder(args).Build().Run();  
15         }  
16  
17         1 reference  
18         public static IHostBuilder CreateHostBuilder(string[] args) =>  
19         {  
20             Host.CreateDefaultBuilder(args)  
21                 .ConfigureWebHostDefaults(webBuilder =>  
22                 {  
23                     webBuilder.UseStartup<Startup>();  
24                 })  
25             ;  
26         }  
27     }  
28 }
```



e) dotnet clean -> dotnet restore -> dotnet build -> dotnet run

```
1 using DotNetEnv;
2 using Microsoft.AspNetCore.Hosting;
3 using Microsoft.Extensions.Hosting;
4
5 namespace WebStorageSample
6 {
7     0 references
8     public class Program
9     {
10         0 references
11         public static void Main(string[] args)
12         {
13             Env.Load(".env");
14             CreateHostBuilder(args).Build().Run();
15         }
16     }
17     1 reference
18     public static IHostBuilder CreateHostBuilder(string[] args) =>
```

Compilación **correcto** con 10 advertencias en 1.9s

Hay actualizaciones de carga de trabajo disponibles. Ejecute "dotnet workload list" para obtener más información.

PS C:\Users\ASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2> dotnet clean

Compilación **realizado correctamente** en 0.4s

PS C:\Users\ASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2>

```
1 using DotNetEnv;
2 using Microsoft.AspNetCore.Hosting;
3 using Microsoft.Extensions.Hosting;
4
5 namespace WebStorageSample
6 {
7     0 references
8     public class Program
9     {
10         0 references
11         public static void Main(string[] args)
12         {
13             Env.Load(".env");
14             CreateHostBuilder(args).Build().Run();
15         }
16     }
17     1 reference
18     public static IHostBuilder CreateHostBuilder(string[] args) =>
```

Hay actualizaciones de carga de trabajo disponibles. Ejecute "dotnet workload list" para obtener más información.

PS C:\Users\ASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2> dotnet clean

Compilación **realizado correctamente** en 0.4s

PS C:\Users\ASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2> dotnet restore

Restaurar **correcto** con 4 advertencias en 0.3s

C:\Users\ASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2\WebStorageSample.csproj : warning NU1903: El paquete "Azure.Identity" 1.4.1 tiene una vulnerabilidad de gravedad alta conocida, https://github.com/advisories/GHSA-5wfx-4wxc-rv27

C:\Users\ASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2\WebStorageSample.csproj : warning NU1902: El paquete "Azure.Identity" 1.4.1 tiene una vulnerabilidad de gravedad moderada conocida, https://github.com/advisories/GHSA-mwv-6r4h-3y9j

C:\Users\ASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2\WebStorageSample.csproj : warning NU1902: El paquete "Azure.Identity" 1.4.1 tiene una vulnerabilidad de gravedad moderada conocida, https://github.com/advisories/GHSA-wxc-855f-jyrv

C:\Users\ASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2\WebStorageSample.csproj : warning NU1902: El paquete "Azure.Storage.Blobs" 12.10.0 tiene una vulnerabilidad de gravedad moderada conocida, https://github.com/advisories/GHSA-64x4-9hc6-r2h6

Compilación **correcto** con 4 advertencias en 0.6s

PS C:\Users\ASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2>



```
1 using DotNetEnv;
2 using Microsoft.AspNetCore.Hosting;
3 using Microsoft.Extensions.Hosting;
4
5 namespace WebStorageSample
6 {
7     0 references
8     public class Program
9     {
10         0 references
11         public static void Main(string[] args)
12         {
13             Env.Load(".env");
14             CreateHostBuilder(args).Build().Run();
15         }
16     }
17     1 reference
18     public static IHostBuilder CreateHostBuilder(string[] args) =>
```

visories/GHSA-Selfx-4wxc-rv27

C:\Users\ASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2\WebStorageSample.csproj : **warni**
ng NU1902: El paquete "Azure.Identity" 1.4.1 tiene una vulnerabilidad de gravedad moderada conocida, <https://github.com/advisories/GHSA-m5vv-6rth-3vj9>

C:\Users\ASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2\WebStorageSample.csproj : **warni**
ng NU1902: El paquete "Azure.Identity" 1.4.1 tiene una vulnerabilidad de gravedad moderada conocida, <https://github.com/advisories/GHSA-m5vv-6rth-3vj9>

C:\Users\ASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2\WebStorageSample.csproj : **warni**
ng NU1902: El paquete "Azure.Storage.Blobs" 12.10.0 tiene una vulnerabilidad de gravedad moderada conocida, <https://github.com/advisories/GHSA-64x4-9hcc-r2h6>

C:\Users\ASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2\StorageHelper.cs(34,17): **warnin**
g CA2200: Si inicia de nuevo la excepción detectada, se cambia la información de la pila. (<https://learn.microsoft.com/dotnet/fundamentals/code-analysis/quality-rules/ca2200>)

C:\Users\ASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2\StorageHelper.cs(66,17): **warnin**
g CA2200: Si inicia de nuevo la excepción detectada, se cambia la información de la pila. (<https://learn.microsoft.com/dotnet/fundamentals/code-analysis/quality-rules/ca2200>)

Compilación **correcto** con **10 advertencias** en 1.55

Hay actualizaciones de carga de trabajo disponibles. Ejecute "dotnet workload list" para obtener más información.
PS C:\Users\ASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2>

```
1 using DotNetEnv;
2 using Microsoft.AspNetCore.Hosting;
3 using Microsoft.Extensions.Hosting;
4
5 namespace WebStorageSample
6 {
7     0 references
8     public class Program
9     {
10         0 references
11         public static void Main(string[] args)
12         {
13             Env.Load(".env");
14             CreateHostBuilder(args).Build().Run();
15         }
16     }
17     1 reference
18     public static IHostBuilder CreateHostBuilder(string[] args) =>
```

NU1902: El paquete "Azure.Identity" 1.4.1 tiene una vulnerabilidad de gravedad moderada conocida, <https://github.com/advisories/GHSA-m5vv-6rth-3vj9>

C:\Users\ASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2\WebStorageSample.csproj : **warning N**
U1902: El paquete "Azure.Storage.Blobs" 12.10.0 tiene una vulnerabilidad de gravedad moderada conocida, <https://github.com/advisories/GHSA-64x4-9hcc-r2h6>

warn: Microsoft.AspNetCore.Server.Kestrel.Core.KestrelServer[8]
The ASP.NET Core developer certificate is not trusted. For information about trusting the ASP.NET Core developer certificate, see <https://aka.ms/aspnet/https-trust-dev-cert>

info: Microsoft.Hosting.Lifetime[14]
Now listening on: <https://localhost:5001>

info: Microsoft.Hosting.Lifetime[14]
Now listening on: <http://localhost:5000>

info: Microsoft.Hosting.Lifetime[0]
Application started. Press Ctrl+C to shut down.

info: Microsoft.Hosting.Lifetime[0]
Hosting environment: Development

info: Microsoft.Hosting.Lifetime[0]
Content root path: C:\Users\ASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2

info: Microsoft.Hosting.Lifetime[0]
Application is shutting down...

PS C:\Users\ASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2>



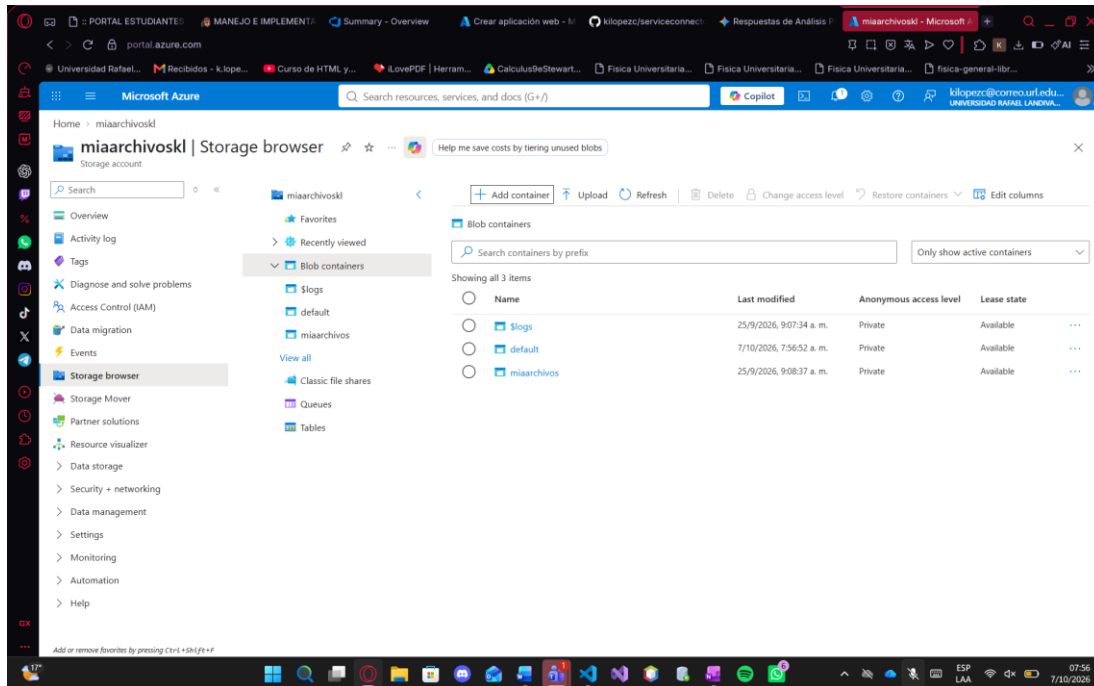
4. Crear recursos en Azure

En azure portal site, cree o utilice un Resource Group, una Storage Account con Blob Storage y un contenedor. El contenedor deberá tener al menos tres archivos de prueba, Puede utilizar el creado en semana 13.

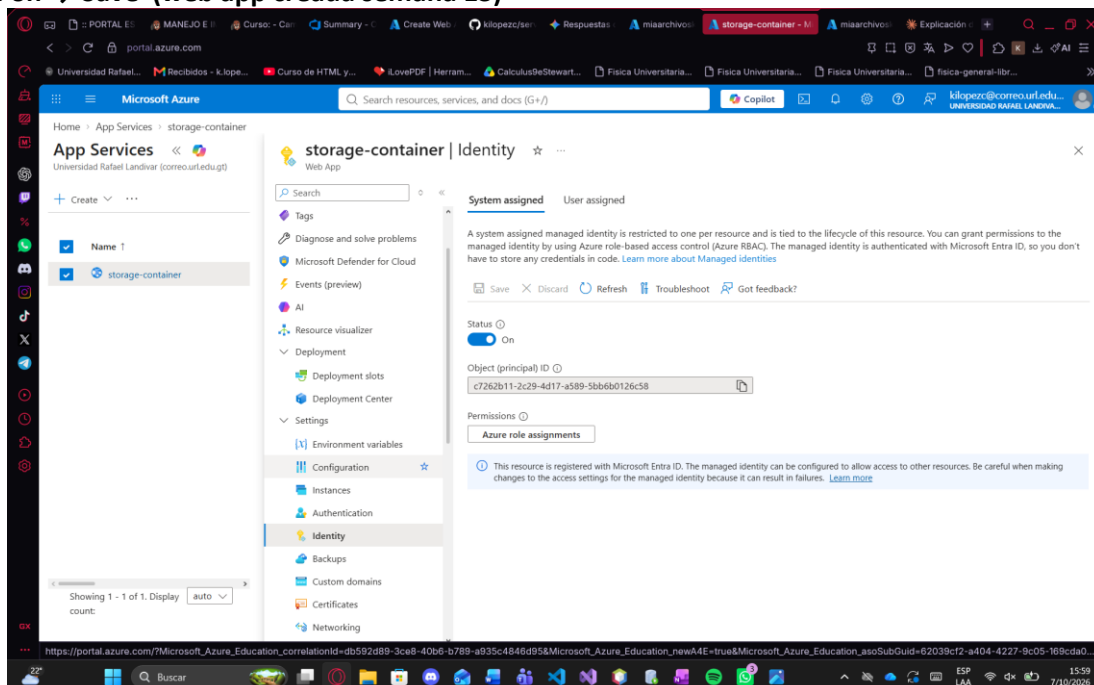
5. Conectar App Service con Blob Storage

Configure la conexión entre App Service y Blob Storage mediante Service Connector utilizando identidad administrada. Verifique que la identidad tenga los permisos necesarios sobre los blobs.

- Azure Portal → **Storage accounts** → tu Storage Account → Data storage → Containers
- Crear contenedor "default"

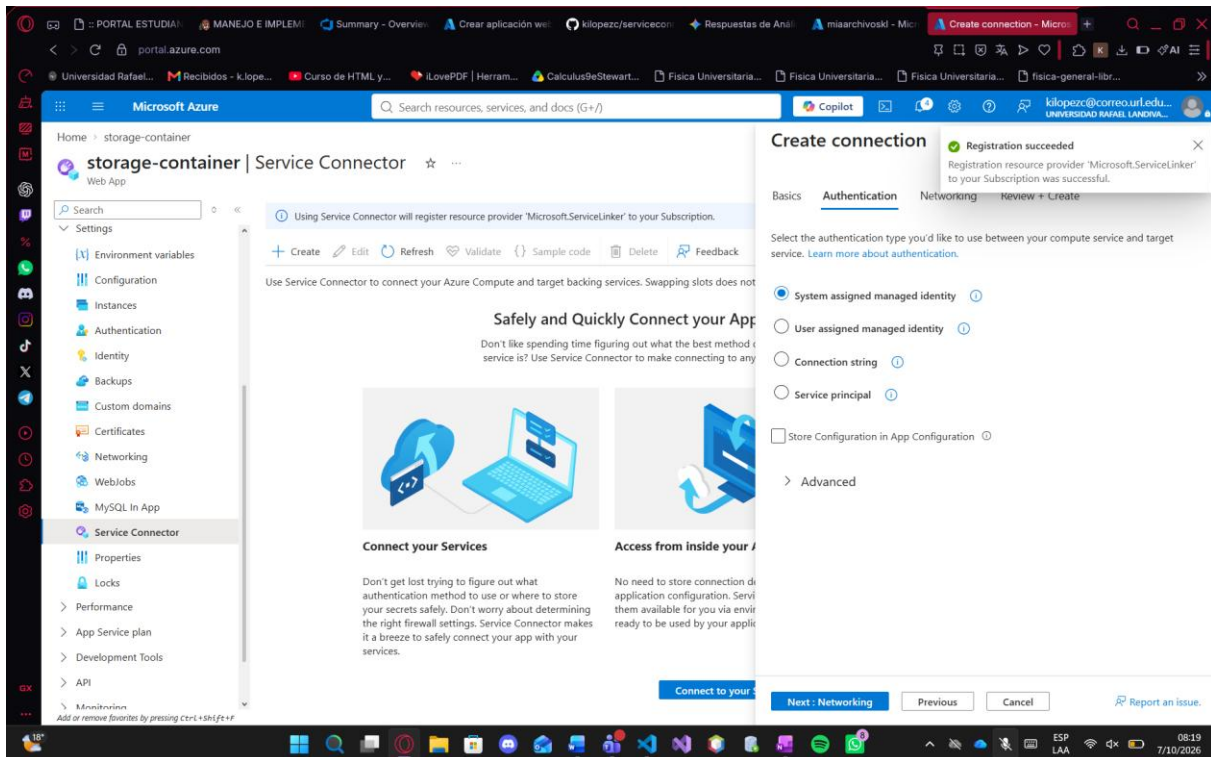
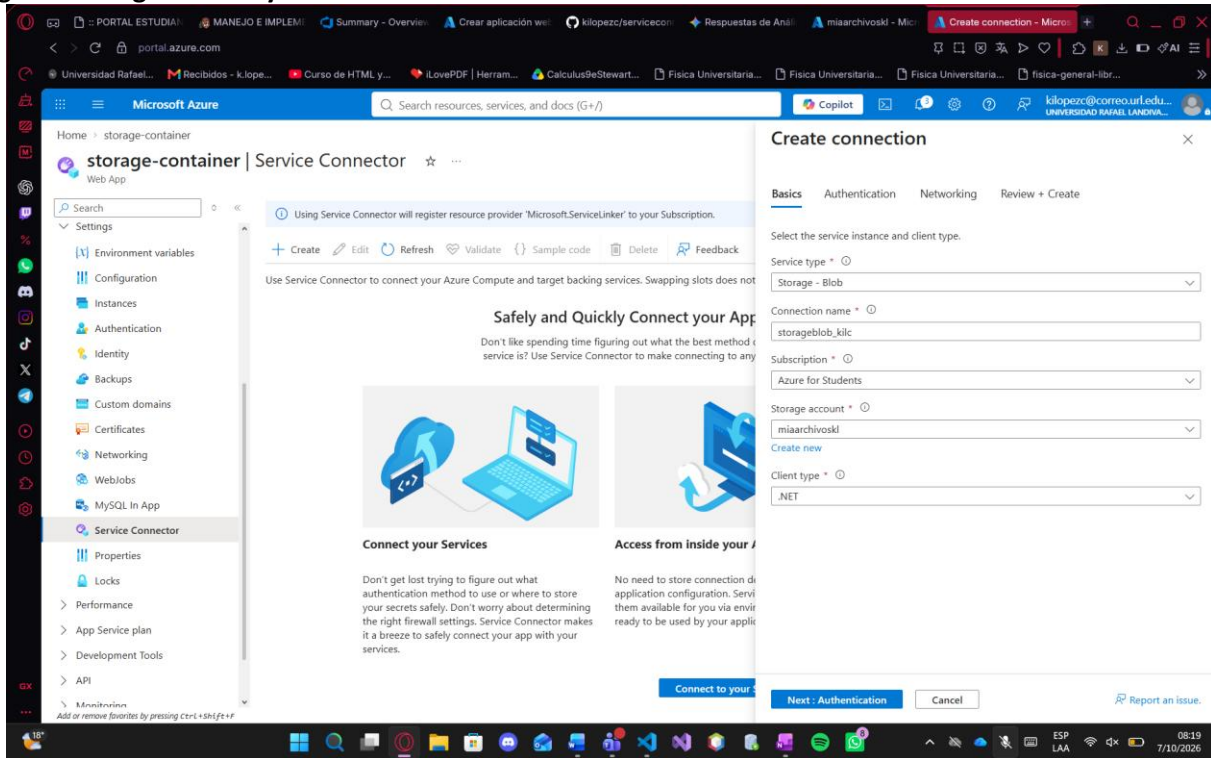


- Activa identidad administrada en App Service: **Azure Portal** → **App Service** → **Settings** → **Identity** → **System assigned: on** → **Save** (web app creada semana 13)





- d) Configurar la conexión entre App Service y Blob Storage: **App Service** → **Settings** → **Service Connector** → **Create** → **Seleccionar "Storage Blob"** → **Seleccionar Storage Account [tu_blob_name]** → **Authentication "System-assigned managed identity"** → **Review + Create**.





Microsoft Azure portal interface showing the 'Create connection' wizard for a Service Connector. The wizard is in the 'Review + Create' step, showing details like Connection name (storageblob_kilc), Subscription (Azure for Students), Client type (.NET), Service type (Storage - Blob), and Storage account (miaarchivoskl). The 'Validation passed' message is visible.

Microsoft Azure portal interface showing the 'Service Connector' page. The page displays a table with columns: Service type, Resource name, Connection name, and Connection status. The table shows one entry: Storage - Blob, miaarchivoskl, storageblob_kilc, and a status icon.

Service type	Resource name	Connection name	Connection status
Storage - Blob	miaarchivoskl	storageblob_kilc	



e) Reemplazar storage blob endpoint y container nombre en .env file

Storage Account → Settings → Endpoints → Blob service

The screenshot displays the Microsoft Azure portal interface. On the left, the 'Storage center | Blob Storage' section is visible, with a list of storage accounts including 'miaarchivoskl'. The main pane shows the 'Endpoints' settings for the 'miaarchivoskl' storage account. The 'Blob service' endpoint is selected, displaying its resource ID and URL. The 'Provisioning state' is 'Succeeded'.

Service	Resource ID	Endpoint URL
Blob service	/subscriptions/62039cf2-a404-4227-9c05-169cda07...	https://miaarchivoskl.blob.core.windows.net/
File service	/subscriptions/62039cf2-a404-4227-9c05-169cda07...	https://miaarchivoskl.file.core.windows.net/
Queue service	/subscriptions/62039cf2-a404-4227-9c05-169cda07...	https://miaarchivoskl.queue.core.windows.net/
Table service	/subscriptions/62039cf2-a404-4227-9c05-169cda07...	https://miaarchivoskl.table.core.windows.net/
Data Lake Storage	/subscriptions/62039cf2-a404-4227-9c05-169cda07...	



6. Probar el acceso al almacenamiento:

Ejecute la aplicación local y compruebe que puede leer los blobs. Realice al menos una operación de carga de archivo y verifique desde Azure Portal que el archivo llegó al contenedor. Visual Studio -> dotnet run

- Verifique que Azure CLI esté instalado ejecutando en la terminal de VS Code: `az --version`
- Inicie sesión en Azure: `az login`
- Verifique la cuenta de Azure activa: `az account show`

```
PS C:\Users\Asus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2> az account show --query "{subscriptionId, user:user.name}" -o table
>> az storage account show --name "tu_storage" --query "{(id:id, resourceGroup:resourceGroup)}" -o table
>> az storage account show --name "tu_storage" --query "{(id:id, resourceGroup:resourceGroup)}" -o table
>> az storage account show --name "miaarchivoskl" --query "{(id:id, resourceGroup:resourceGroup)}" -o table
```

- Verifique que pueda obtener un token para Azure Storage: `az account get-access-token --resource https://storage.azure.com/`

```
PS C:\Users\Asus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2> az account get-access-token --resource https://storage.azure.com/
{
  "accessToken": "3h0eGp0M5G0R0U2c1m0PS2h0e0X0k0a0z0F0V0Z0T0K080Q0M0Z0I0X0G0I0Z0k0a0z0H0I0w0V0T0j0b0t0B0Z0C0d0C0h0C0p0Z0C0G0j0b0Y0A0g0k0l0Th0k0z0t0H0x0s0L0Y0m0L0I0y0Z0I0P0d0m0Z0I0H0I0s0w0c0k0w0Y0j0g0P0c0s0m0h0d0s0p0w0d0I0j0t0E0S0Q0v0g0H0T0E0J0t0E0B0L0C0h0a0z0B0a0m0d0I0j0t0S0Q0T0V0S0C0U0P0J0M0P0I0S0Q0I0C0p0R0S0C0G0I0Z0X0I0C0p0G0I0Z0I0I0I0Y0D0a0Q0P0G0I0H0C0P0I0Y0g0d0H0I0M0P0a0t0c0I0j0k0D0g0I0b0E0T0S0I0K0F0T0S0V0I0P0S0M0P0A0T0I0S0E0I0E0P0V0A0I0H0U0R0T0P0I0w0L0k0j0I0Y0Z0Y0K0I0Y0C0M0T0V0K0a0w0Y0L0I0J0T0T0M0Z0D0H0K0H0P0I0a0C0H0P0Z0C0G0j0b0Y0A0Z0B0R0C0z0P0R0U0I0C0Y0a0C0I0j0e0Q0V0Z0Q0A0J0R0L0X0c0j0P0M0T0I0C0Q0F0z0h0K0U0I0V0U0M0I0a0b0C0Q0H0B0P0I0d0h0a0Q0g0B0R0L0I0s0t0I0j0C0I0f0a0w0Z0I0Z0B0Z0h0I0V0I0S0I0P0Z0C0G0I0j0w0T0I0Z0H0U0M0Z0G0I0T0I0d0H0a0z0J0B0L0T0W0Q0I0C0Z0H0Y0S0I0H0Y0I0I0J0B0K0I0a0G0a0Q0X0P0H0H0T0S0C0Y0J0J0S0P0R0C0M0H0E0Y0M0F0Z0K0R0E0Q0I0C0R0M0Q0I0I0w0J0c0M0T0S0C0Z0A0J0I0Q0W0V0T0J0Z0S0Z0J0I0P0Q0Z0Z0J0M0I0C0I0B0I0d0V0f0E0T0I0a0I0K0a0Z0G0Z0G0a0C0I0Y0S0I0C0w0Z0P0I0I0d0B0I0j0a0Z0I0S0H0I0E0M0V0Z0Y0c0w0I0y0C0S0I0Z0H0Z0Q0I0C0I0d0I0I0J0c0M0I0W0J0V0E0M0H0P0C0P0I0B0I0U0d0Y0I0j0I0S0w0I0d0C0I0Z0F0J0F0w0Q0I0I0I0T0H0C0B0H0P0V0A0d0I0A0Q0I0I0I0K0A0Z0V0S0I0W0Z0Q0R0T0Y0I0T0B0I0I0J0H0I0W0I0K0I0D0H0Z0J0I0C0B0H0F0Z0K0I0j0a0I0H0A0Z0B0I0A0I0U0d0S0F0U0I0U0B0E0U0M0I0d0M0H0U0B0V0R0Y0F0V0U0I0K0E0S0Y0J0M0F0M0E0j0J0F0I0d0e0G0Z0Z0I0K0M0V0I0j0I0S0Y0C0S0I0H0C0I0Z0d0I0Z0M0I0j0I0Y0A0Z0I0H0C0H0E0W0Z0P0I0U0P0I0F0d0C0V0Z0P0J0a0R0S0a0G0V0C0Z0Y0U0M0F0C0D0a0P0I0U0d0Z0S0Z0I0C0B0I0j0S0V0K0G0Z0I0B0V0P0J0I0P0K0F0J0d0S0Z0U0M0V0Z0M0D0E0W0C0a0I0R0D0F0Z0C0C0M0K0I0J0S0B0A0P0P0R0I0B0C0F0B0a0Q0I0Q0I0Y0A0C0I0B0I0N0R0L0Y0D0P0M0P0X0I0D0K0I0I0I0d0B0H0M0E0P0S0W0I0G0a0O0P0E0I0P0Z0L0I0Z0P0G0S0R0H0M0I0a0Z0S0F0B0I0K0I0I0Y0J0V0I0W0Z0I0M0I0R0V0I0J0Y0V0J0C0I0H0Z0A0M0C0I0M0I0P07a0p0I0Z0C0D0K0R0C0I0H0E0A0Z0V0S0S0E0a0Q",
  "expiresOn": "2026-10-07 18:25:43.4000000",
  "expiresOnT": "1791419143",
  "subscription": "62039cf2-a044-4227-9c05-169cd07942b",
  "tenant": "0f78540d-3eec-43af-b56a-6f7b042d0c9a",
  "tokenType": "bearer"
}
```



e) Si se muestra un accessToken la autenticación esta correcta, ejecutar -> dotnet run

f) Si hay error en acceso, ejecutar en terminal vs code: az role assignment create --assignee

"TU_CORREO_DE_AZURE" --role "Storage Blob Data Contributor" --scope

"/subscriptions/TU_SUBSCRIPTION_ID/resourceGroups/TU_RESOURCE_GROUP/providers/Microsoft.Storage/storageAccounts/TU_STORAGE"

Obtener info: az account show --query "{subscriptionId:id, user:user.name}" -o table

az storage account show --name "tu_storage" --query "{id:id, resourceGroup:resourceGroup}" -o table

```

PS C:\Users\Asus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2> az account show --query "{subscriptionId:id, user:user.name}" -o table
SubscriptionId      User
-----
62039cf2-a404-4227-9c05-169cda07942b kilopez@correo.url.edu.gt
ResourceGroup
-----
rg-mia-lab

PS C:\Users\Asus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2> az storage account show --name "miaarchivoskl" --query "{id:id, resourceGroup:resourceGroup}" -o table
>>
SubscriptionId      User
-----
62039cf2-a404-4227-9c05-169cda07942b kilopez@correo.url.edu.gt
ResourceGroup
-----
rg-mia-lab

PS C:\Users\Asus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2> az role assignment create --assignee "kilopez@correo.url.edu.gt" --role "Storage Blob Data Contributor" --scope "/subscriptions/62039cf2-a404-4227-9c05-169cda07942b/resourceGroups/rg-mia-lab/providers/Microsoft.Storage/storageAccounts/TU_STORAGE" (ResourceNotFound) The Resource 'Microsoft.Storage/storageAccounts/TU_STORAGE' under resource group 'rg-mia-lab' was not found. For more details please go to https://aka.ms/ARNResourceNotFoundFix
Code: ResourceNotFound
Message: The Resource 'Microsoft.Storage/storageAccounts/TU_STORAGE' under resource group 'rg-mia-lab' was not found. For more details please go to https://aka.ms/ARNResourceNotFoundFix
PS C:\Users\Asus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2> az role assignment create --assignee "kilopez@correo.url.edu.gt" --role "Storage Blob Data Contributor" --scope "/subscriptions/62039cf2-a404-4227-9c05-169cda07942b/resourceGroups/rg-mia-lab/providers/Microsoft.Storage/storageAccounts/miaarchivoskl"

```

```

PS C:\Users\Asus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2> az role assignment create --assignee "kilopez@correo.url.edu.gt" --role "Storage Blob Data Contributor" --scope "/subscriptions/62039cf2-a404-4227-9c05-169cda07942b/resourceGroups/rg-mia-lab/providers/Microsoft.Storage/storageAccounts/miaarchivoskl"
{
  "conditionVersion": null,
  "createdBy": null,
  "createdon": "2026-10-07T23:26:53.521084+00:00",
  "delegatedManagedIdentityResourceId": null,
  "description": null,
  "id": "/subscriptions/62039cf2-a404-4227-9c05-169cda07942b/resourceGroups/rg-mia-lab/providers/Microsoft.Storage/storageAccounts/miaarchivoskl/providers/Microsoft.Authorization/roleAssignments/f28c163c-4256-4481-b379-6087929061ae",
  "name": "f28c163c-4256-4481-b379-6087929061ae",
  "principalId": "c691fe7-15e4-4ab2-b567-9343c55955c4",
  "principalType": "User",
  "resourceGroup": "rg-mia-lab",
  "roleDefinitionId": "/subscriptions/62039cf2-a404-4227-9c05-169cda07942b/providers/Microsoft.Authorization/roleDefinitions/ba92f5bd-2d11-453d-a403-e9600829c9fe",
  "scope": "/subscriptions/62039cf2-a404-4227-9c05-169cda07942b/resourceGroups/rg-mia-lab/providers/Microsoft.Storage/storageAccounts/miaarchivoskl",
  "systemdata": null,
  "type": "Microsoft.Authorization/roleAssignments",
  "updatedBy": "c691fe7-15e4-4ab2-b567-9343c55955c4",
  "updatedon": "2026-10-07T23:26:53.777089+00:00"
}
PS C:\Users\Asus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2>

```



g) Cerrar process ctr +c y logear de nuevo: az logout y az login, Por ultimo dotnet run

```

1 AZURE_STORAGEBLOB_RESOURCEENDPOINT=https://miaarchivos1.blob.core.windows.net/
2 AZURE_STORAGE_CONTAINER_NAME=default
3

serviceconnector-webapp-storageblob-dotnet2 > .env.production
PS C:\Users\Asus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2> az login
PS C:\Users\Asus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2> az login
A web browser has been opened at https://login.microsoftonline.com/organizations/oauth2/v2.0/authorize. Please continue the login in the web browser. If no web browser is available or if the web browser fails to open, use device code flow with 'az login --use-device-code'.

Retrieving tenants and subscriptions for the selection...

[Tenant and subscription selection]

No  Subscription name  Subscription ID  Tenant
-----
[1] * Azure for Students  62039cf2-a404-4227-9c05-169cda07942b  Universidad Rafael Landívar

The default is marked with an *; the default tenant is 'Universidad Rafael Landívar' and subscription is 'Azure for Students' (62039cf2-a404-4227-9c05-169cda07942b).

Select a subscription and tenant (Type a number or Enter for no changes): 1
  
```

Nota: no olvidar subir cambios a repo github fork

7. Crear Pipeline Yml

En la raíz de tu fork de GitHub crear archivo "azure-pipelines.yml"

Colocar en el archivo lo siguiente:

trigger:

- main

pool:

vmImage: 'windows-latest'

variables:

buildConfiguration: 'Release'

dotnetVersion: '9.x'

Completar posteriormente

azureSubscription: 'NOMBRE-DE-TU-SERVICE-CONNECTION'

appName: 'NOMBRE-DE-TU-APP-SERVICE'

steps:

1. Instalar .NET 9

- task: UseDotNet@2

displayName: 'Instalar .NET 9 SDK'

inputs:

packageType: 'sdk'



2. Restaurar dependencias

```
- task: DotNetCoreCLI@2
  displayName: 'Restaurar dependencias'
  inputs:
    command: 'restore'
    projects: '**/*.csproj'
```

3. Configurar ambiente de producción

```
- powershell: |
  Copy-Item ".env.production" ".env" -Force
  displayName: 'Configurar .env de producción'
```

4. Compilar

```
- task: DotNetCoreCLI@2
  displayName: 'Compilar aplicación'
  inputs:
    command: 'build'
    projects: '**/*.csproj'
    arguments: '--configuration $(buildConfiguration) --no-restore'
```

5. Publicar

```
- task: DotNetCoreCLI@2
  displayName: 'Publicar aplicación'
  inputs:
    command: 'publish'
    projects: '**/*.csproj'
    arguments: '--configuration $(buildConfiguration) --output $(Build.ArtifactStagingDirectory)/publish --no-build'
    zipAfterPublish: true
```

6. Desplegar en Azure App Service

```
- task: AzureWebApp@1
  displayName: 'Desplegar en Azure App Service'
  inputs:
    azureSubscription: '$(azureSubscription)'
    appType: 'webApp'
    appName: '$(appName)'
    package: '$(Build.ArtifactStagingDirectory)/publish/**/*.zip'
```



The screenshot shows the GitHub web interface for the repository `serviceconnector-webapp-storageblob-dotnet2`. The file `azure-pipelines.yml` is open in the editor. The left sidebar shows the file explorer with the `main` branch selected. The right sidebar shows the file's metadata and a "Commit changes" button. The main editor area displays the YAML content of the pipeline.

```
1 trigger:
2   - main
3
4 pool:
5   vmImage: 'windows-latest'
6
7 variables:
8   buildConfiguration: 'Release'
9   dotnetVersion: '9.x'
10
11 # Completar posteriormente
12 azureSubscription: 'NOMBRE-DE-TU-SERVICE-CONNECTION'
13 appName: 'NOMBRE-DE-TU-APP-SERVICE'
14
15 steps:
16
17 # 1. Instalar .NET 9
18 - task: UseDotNet@2
19   displayName: 'Instalar .NET 9 SDK'
20   inputs:
21     packageType: 'sdk'
22     version: '$(dotnetVersion)'
23
24 # 2. Restaurar dependencias
25 - task: DotNetCoreCLI@2
26   displayName: 'Restaurar dependencias'
27   inputs:
28     command: 'restore'
```

The screenshot shows the same GitHub repository, but now the file `azure-pipelines.yml` is selected in the file explorer on the left. The main editor area shows the file's metadata, including the commit hash `5910c19` and the file size `60 lines (50 loc) - 1.43 KB`. The file content is displayed in the main editor area.

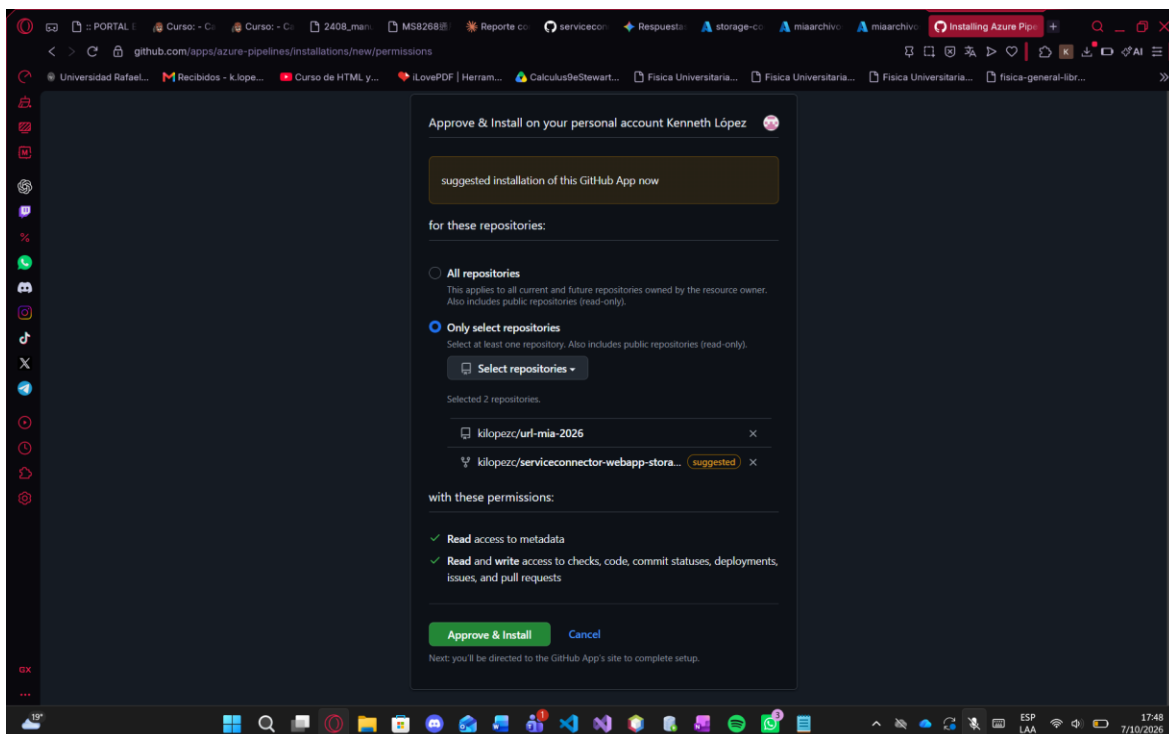
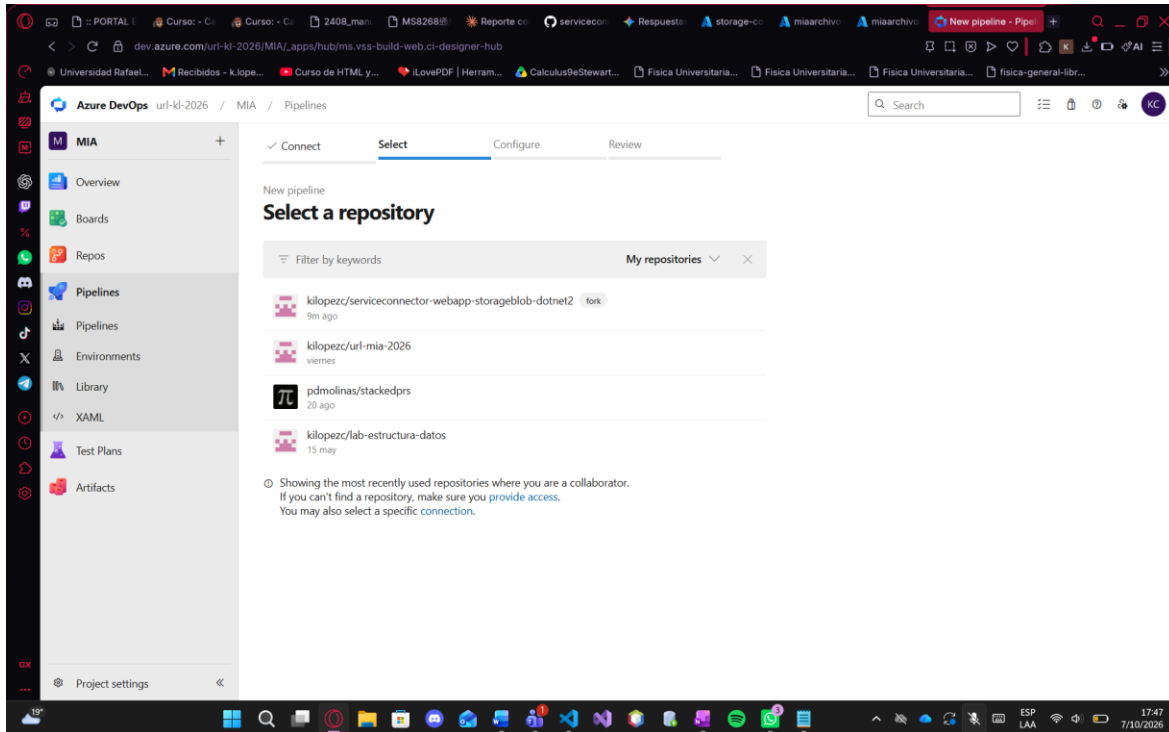
```
1 trigger:
2   - main
3
4 pool:
5   vmImage: 'windows-latest'
6
7 variables:
8   buildConfiguration: 'Release'
9   dotnetVersion: '9.x'
10
11 # Completar posteriormente
12 azureSubscription: 'NOMBRE-DE-TU-SERVICE-CONNECTION'
13 appName: 'NOMBRE-DE-TU-APP-SERVICE'
14
15 steps:
16
17 # 1. Instalar .NET 9
18 - task: UseDotNet@2
19   displayName: 'Instalar .NET 9 SDK'
20   inputs:
21     packageType: 'sdk'
22     version: '$(dotnetVersion)'
23
24 # 2. Restaurar dependencias
25 - task: DotNetCoreCLI@2
26   displayName: 'Restaurar dependencias'
27   inputs:
28     command: 'restore'
29     projects: '**/*.csproj'
```

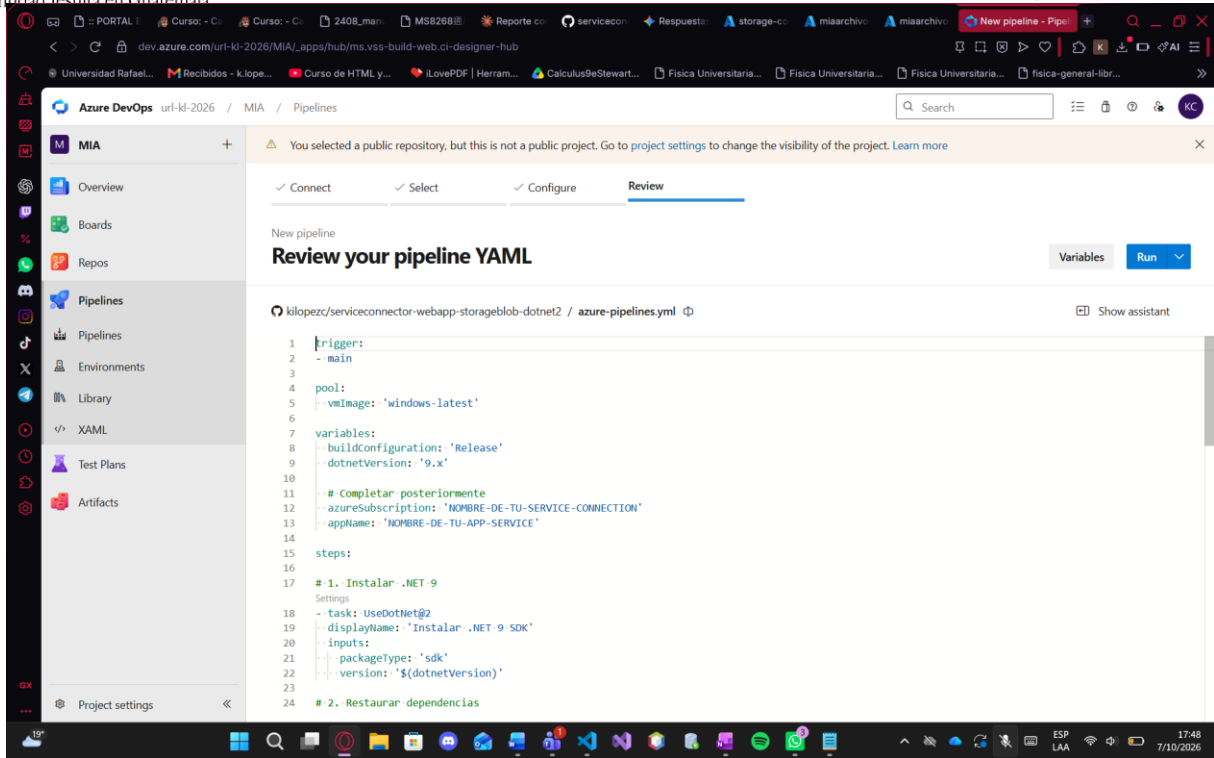


8. Crear pipeline CI/CD

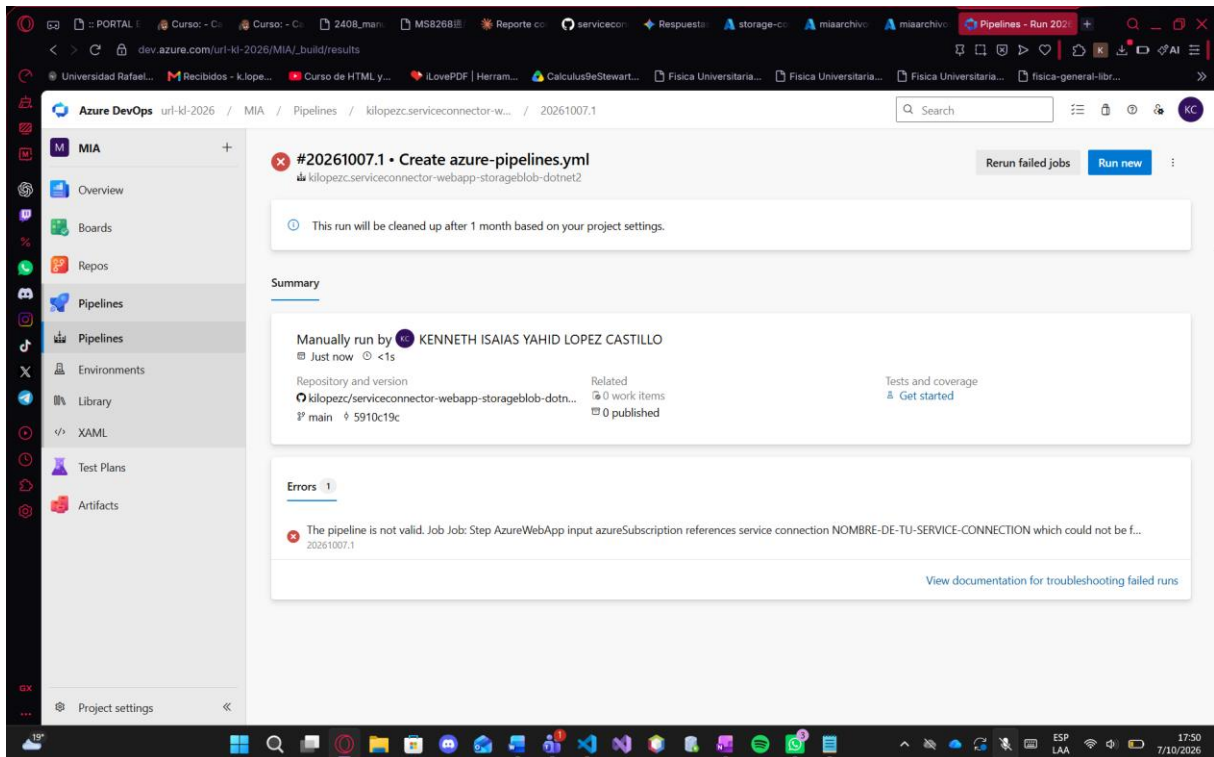
Cree un pipeline YAML que restaure dependencias, compile, publique y despliegue la aplicación en Azure App Service.

- Conectarse a Azure DevOps son misma cuenta que azure portal -> <https://aex.dev.azure.com/me>
- Crear pipeline: **En Azure devOps -> Menu izquierda -> pipelines -> Create Pipeline -> Seleccionar Github -> Autoriza Azure DevOps para acceder a GitHub -> seleccionar repositorio que hizo fork -> seleccionar Existing Azure Pipelines YAML file -> Branch: main Path: /azure-pipelines.yml**



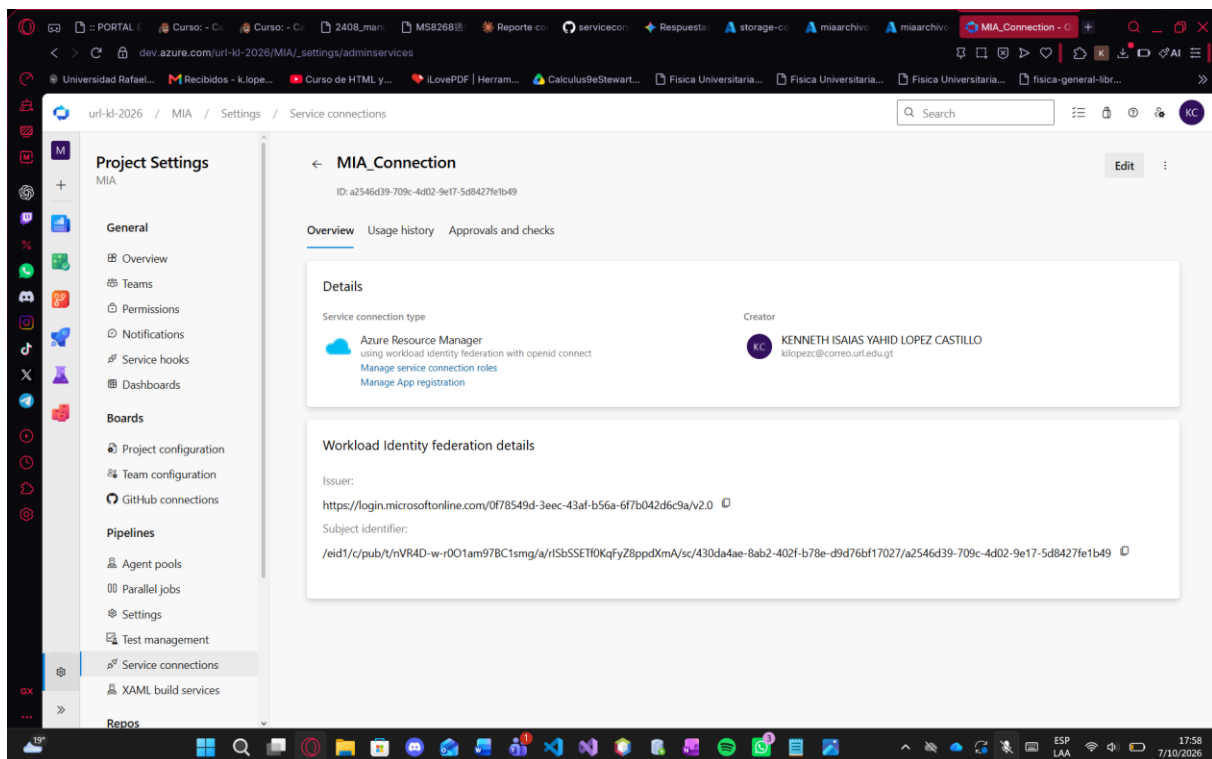
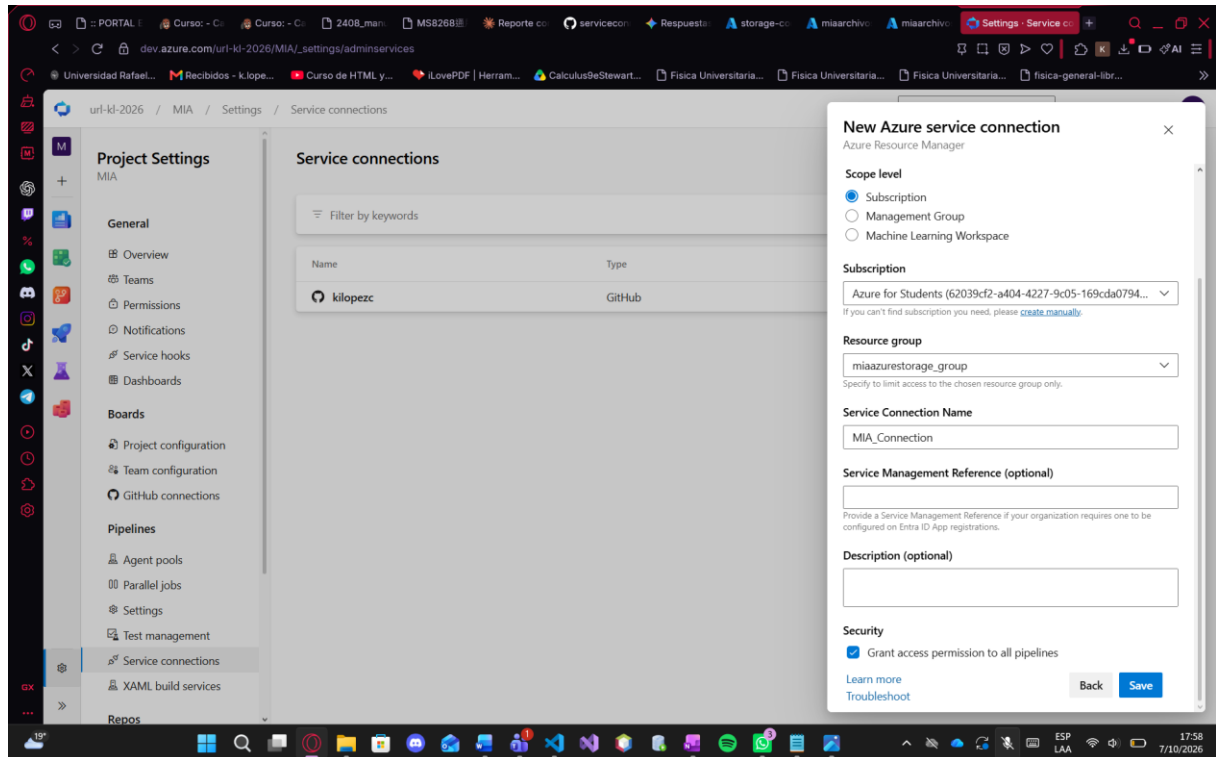


c) Probar seleccionando “run”





- d) Conectar Azure DevOps con Azure: **Project Settings** → **Service connections** → **New service connection** -> **Azure Resource Manager** -> luego, **Workload identity federation** -> **Selecciona tu suscripción de Azure** -> y **resource group** donde estan tus recursos -> **Añadir nombre "MIA_Connection"** -> **Seleccionar "grant access permission to all pipelines"** -> **Save**





8. Conectar Pipeline con App

a) Obtener nombre app: En Azure Portal App Services → tu aplicación → Copia nombre app

The screenshot shows the 'Storage account' page for 'miaarchivoskl' in the Microsoft Azure Portal. The left sidebar contains navigation options like Overview, Activity log, Tags, Diagnose and solve problems, Access Control (IAM), Data migration, Events, Storage browser, Storage Mover, Partner solutions, Resource visualizer, Data storage, Containers, Classic file shares, Queues, Tables, Security + networking, Networking, and Front Door and CDN. The main content area shows the 'Essentials' section with details such as Resource group (rg-mia-lab), Location (westus), Subscription (Azure for Students), Subscription ID (62039cf2-a404-4227-9c05-169cda07942b), Disk state (Available), and Tags (Add tags). Below this, the 'Properties' section is visible, showing 'Blob service' settings like Hierarchical namespace (Disabled), Default access tier (Hot), Blob anonymous access (Disabled), Blob soft delete (Enabled (7 days)), Container soft delete (Enabled (7 days)), Versioning (Disabled), Change feed (Disabled), NFS v3 (Disabled), Allow cross-tenant replication (Disabled), and Storage tasks assignments (None). The 'Security' section shows 'Require secure transfer for REST API operations' (Enabled), 'Storage account key access' (Enabled), 'Minimum TLS version' (Version 1.2), and 'Infrastructure encryption' (Disabled). The 'Networking' section shows 'Public network access' (Enabled), 'Public network access scope' (Enable from all networks), 'Private endpoint connections' (0), 'Network routing' (Microsoft network routing), and 'Endpoint type' (Standard).

The screenshot shows the 'storage-container' Web App page in the Microsoft Azure Portal. The left sidebar contains navigation options like Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Microsoft Defender for Cloud, Events (preview), AI, Resource visualizer, Deployment, Settings, Environment variables, Configuration, Instances, Authentication, Identity, Backups, Custom domains, and Certificates. The main content area shows the 'Essentials' section with details such as Resource group (miaazurestorage_group), Status (Running), Location (West US), Subscription (Azure for Students), Subscription ID (62039cf2-a404-4227-9c05-169cda07942b), Tags (Add tags), Default domain (storage-container-hjh5e6e2gcd6bsgz.westus-01.azurewebsites.net), App Service Plan (ASP-miaazurestoragegroup-a14b), Operating System (Windows), and Health Check (Not Configured). Below this, the 'Properties' section is visible, showing 'Web app' settings like Name (storage-container), Publishing model (Code), Runtime Stack (.NET - v9.0), and Deployment Center settings like Deployment logs (View logs), Last deployment (Loading deployments...), and Deployment provider (VSTRM). The 'Domains' section shows 'Default domain' (storage-container-hjh5e6e2gcd6bsgz.westus-01.azurewebsites.net) and 'Custom domain' (Add custom domain). The 'Application Insights' section shows 'Name' (Enable Application Insights) and 'Networking' (Inbound IP addresses: 2603:1030:a07:402::af).



b) en github en tu "azure-pipelines.yml" file busca: azureSubscription y appName, cambiar por valores reales.

azureSubscription: TU_Service_Connection_name

appName: Tu_app_name

The screenshot shows the Visual Studio Code interface. The Explorer pane on the left shows the file structure of the 'WebStorageSample.csproj' project. The main editor displays the 'azure-pipelines.yml' file with the following content:

```
1 trigger:
2   - main
3
4 pool:
5   vmImage: 'windows-latest'
6
7 variables:
8   buildConfiguration: 'Release'
9   dotnetVersion: '9.x'
10
11 # Completar posteriormente
12 azureSubscription: 'MIA_Connection'
13 appName: 'storage-container'
14
15 steps:
16
17   # 1. Instalar .NET 9
18   - task: UseDotNet@2
19     displayName: 'Instalar .NET 9 SDK'
```

The terminal at the bottom shows the output of running 'git pull origin' and 'git status', indicating that the branch is up to date.

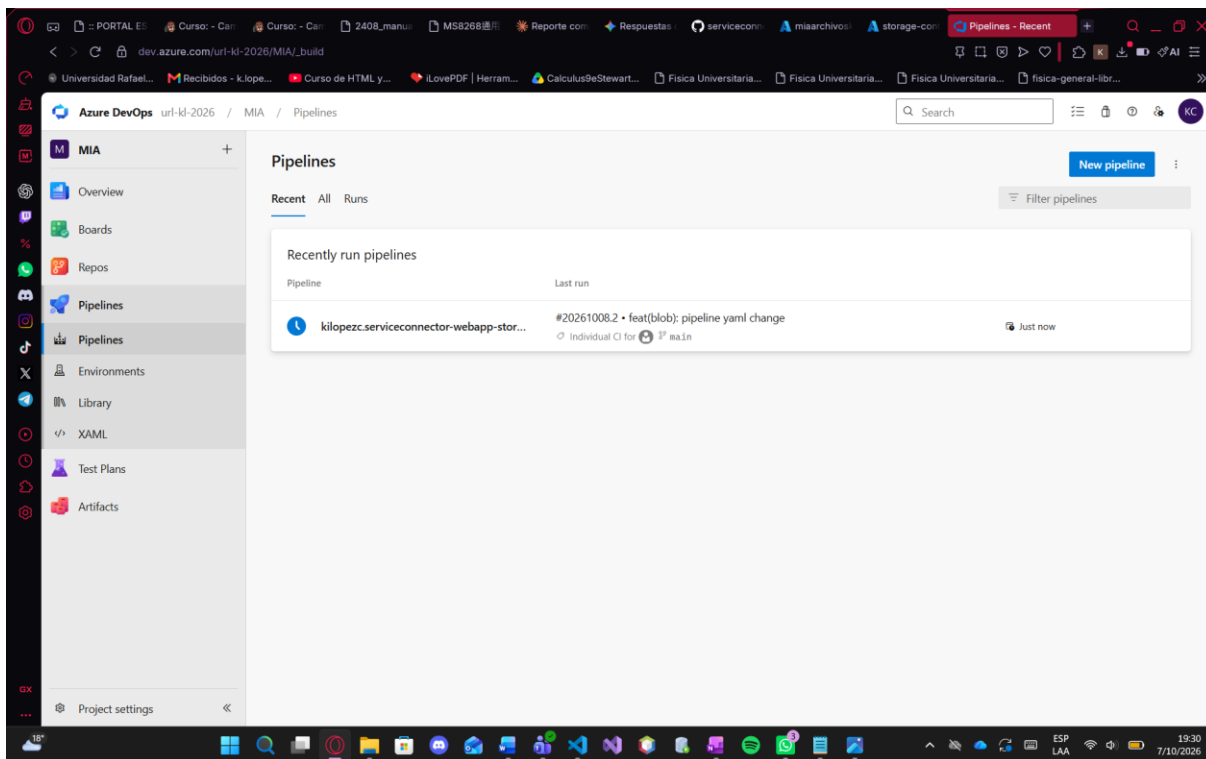
c) Al hacer commit, verificar el pipeline se ejecuta automáticamente

The screenshot shows the GitHub repository page for 'serviceconnector-webapp-storageblob-dotnet2'. The 'Files' tab is selected, and the 'azure-pipelines.yml' file is open. The file content is the same as in the previous screenshot. The commit history shows a recent commit by 'kenneth20' with the message 'feature: agregar pipeline'.



The screenshot shows the Visual Studio Code interface with the following components:

- Explorer:** Displays the file structure of the project, including files like `azure-pipelines.yml`, `CHANGELOG.md`, `Const.cs`, `CONTRIBUTING.md`, `LICENSE.md`, `Program.cs`, `README.md`, `Startup.cs`, and `StorageHelper.cs`.
- azure-pipelines.yml:** The configuration file for the pipeline, showing variables for build configuration and steps for installing .NET 9.0 and using the UseDotNet@2 task.
- Terminal:** Displays the output of the pipeline run, including the command `git add azure-pipelines.yml`, `git commit -m "feat(blob): pipeline yaml change"`, and `git push`. It also shows the output of the `git push` command, indicating that the pipeline was successfully pushed to the repository.
- Chat:** A sidebar on the right with a "Build with Agent" button and a "Generate Agent Instructions to onboard AI onto your codebase" button.





The screenshot shows the Azure DevOps Pipelines interface for the pipeline named 'kilopecz.serviceconnector-webapp-storageblob-dotnet2'. The left sidebar shows the project structure with 'Pipelines' selected. The main area displays a table of pipeline runs.

Description	Stages	Run Status	Time
#20261008.2 • feat(blob): pipeline yaml change Individual CI for main 763788e4	1 stage (failed)	Failed	Just now 1m 3s
#20261008.1 • feature: agregar pipeline Individual CI for main aa9af24e	1 stage (succeeded)	Succeeded	1h ago 3m 15s
#20261007.1 • Create azure-pipelines.yml Manually run by main 5918c19c	1 stage (failed)	Failed	2h ago <1s

The screenshot shows a web browser displaying the 'WebStorageSample' application. The page has a header with 'WebStorageSample', 'Home', and 'Privacy' links. The main content area displays the message 'Hello Service Connector! UTC Now: 10/8/2026 1:35:17 AM +00:00.' and a link to 'Learn about building Web apps with ASP.NET Core.' The footer shows '© 2021 - WebStorageSample - Privacy'.



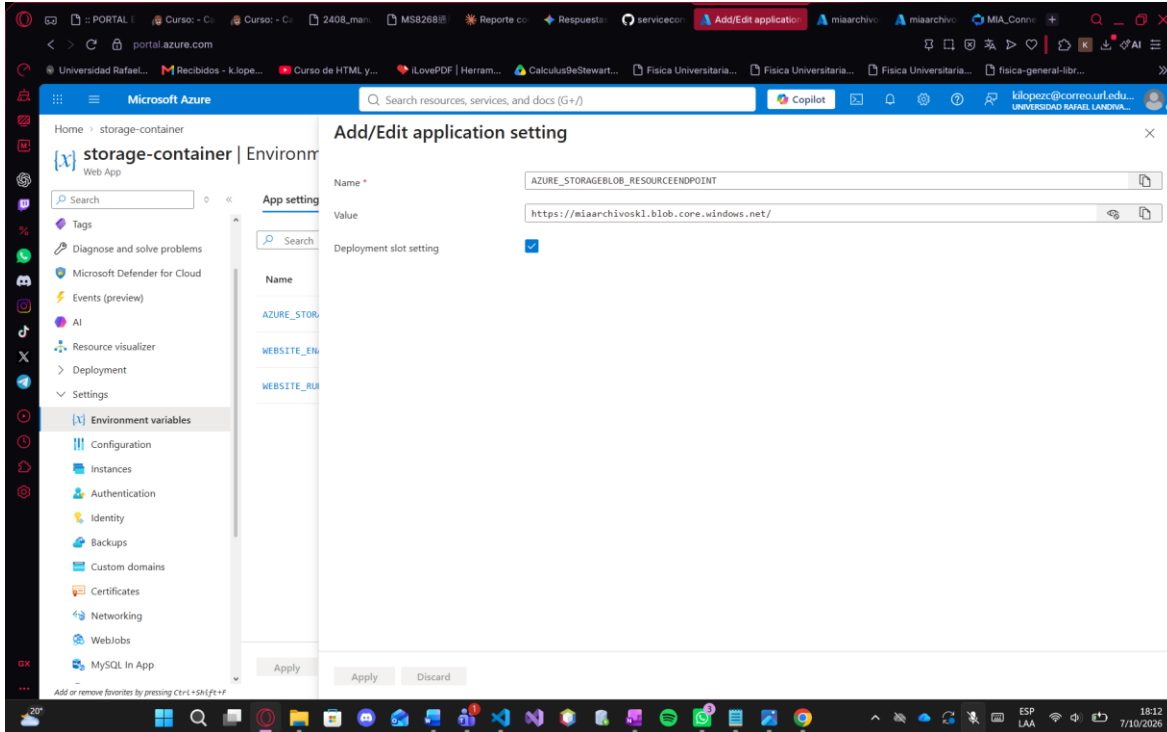
9. Validar la solución

Compruebe desde la URL pública que la aplicación funciona y que el acceso al Blob Storage continúa operativo después del despliegue.

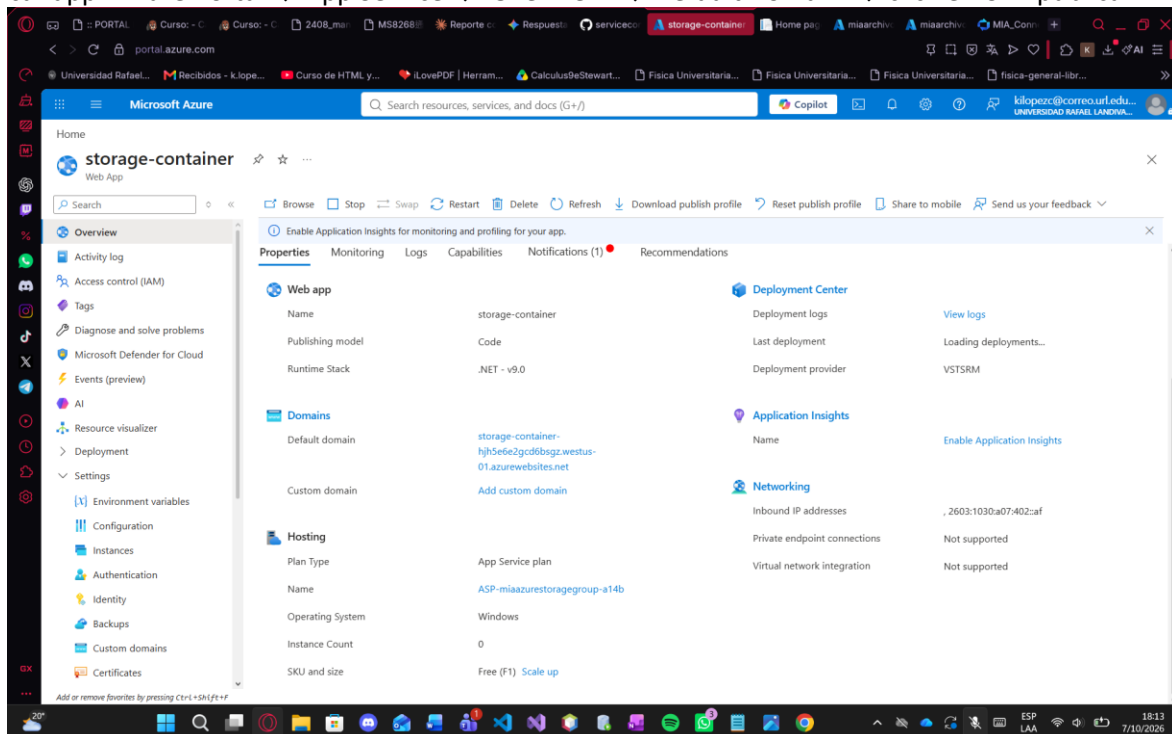
- a) acceso de la aplicación al Blob: Azure Portal → App Service → Settings → Environment variables → App settings
-> colocar AZURE_STORAGEBLOB_RESOURCEENDPOINT

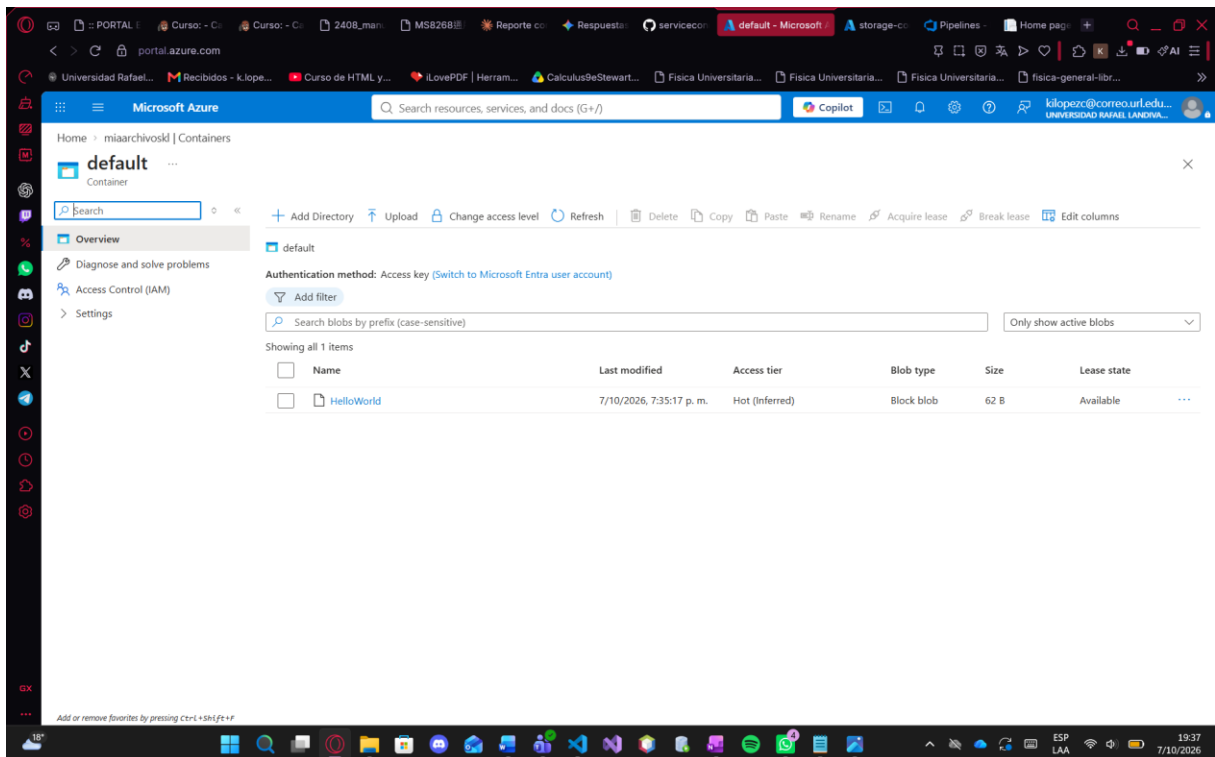
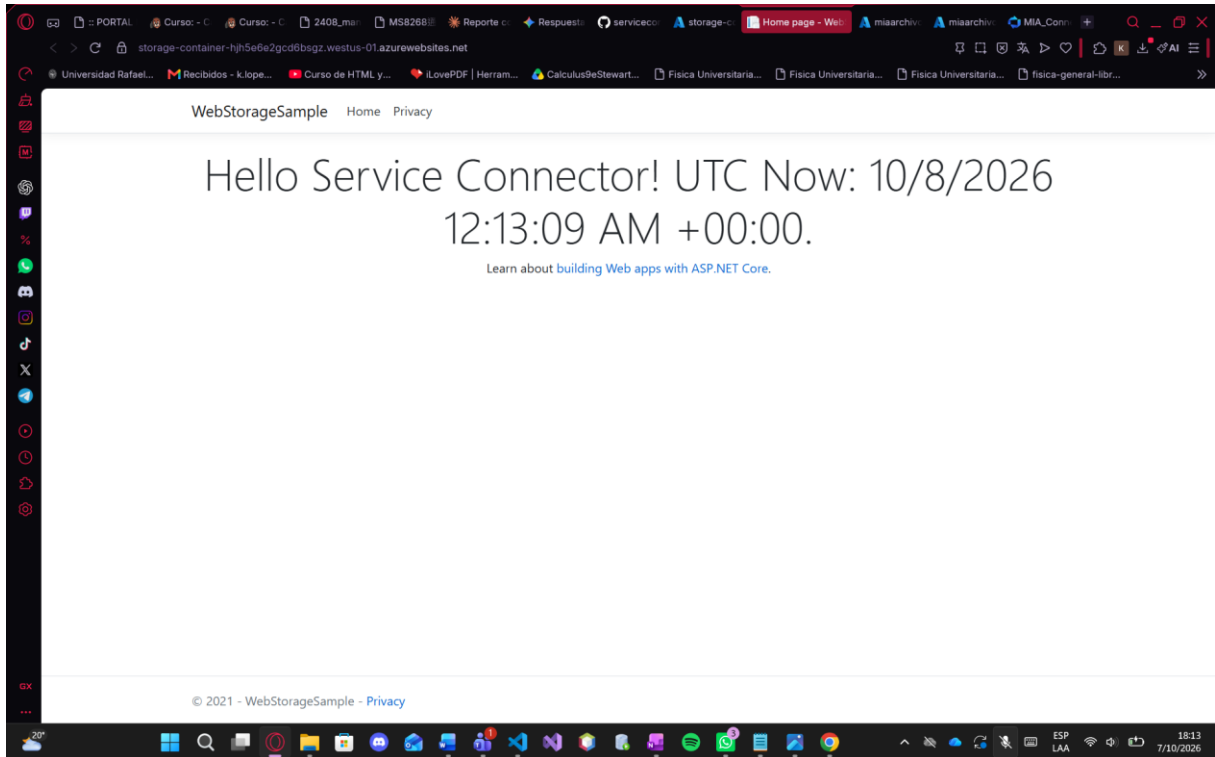
https://tu_storage.blob.core.windows.net/

(si existe obviar)



- b) Verificar app: Azure Portal → App Service → Overview → Default Domain → click en URL publica







c) En la app publicada realizar cambios en archivos

```
1 @page
2
3 @model WebStorageSample.Pages.IndexModel
4
5 @{
6     ViewData[Index, "Title"] = "Azure Blob Storage";
7 }
8
9 <div class="text-center">
10
11     <h1 class="display-5">Azure Blob Storage</h1>
12
13     <p>Seleccione una imagen para almacenarla en Azure Blob Storage.</p>
14
15     <form method="post" enctype="multipart/form-data">
16
17         <div class="mb-3">
18             <input type="file"
19                 name="image"
20                 accept="image/*"
21                 class="form-control" />
22
23             <button type="submit" class="btn btn-primary">
24                 Subir imagen
25             </button>
26
27         </div>
28
29         @if (!string.IsNullOrEmpty(Model.ImageBase64))
30         {
31             <div class="mt-4">
32
33             </div>
34
35         }
36     </form>
37 </div>
```

```
1 using System;
2 using System.Collections.Generic;
3 using System.Diagnostics;
4 using System.Linq;
5 using System.Threading.Tasks;
6 using Microsoft.AspNetCore.Mvc;
7 using Microsoft.AspNetCore.Mvc.RazorPages;
8 using Microsoft.Extensions.Logging;
9
10 namespace WebStorageSample.Pages
11 {
12     [ResponseCache(Duration = 0, Location = ResponseCacheLocation.None, NoStore = true)]
13     public class ErrorModel : PageModel
14     {
15         public string RequestId { get; set; }
16
17         public bool ShowRequestId => !string.IsNullOrEmpty(RequestId);
18
19         private readonly ILogger<ErrorModel> _logger;
20
21         public ErrorModel(ILogger<ErrorModel> logger)
22         {
23             _logger = logger;
24         }
25
26         public void OnGet()
27         {
28             RequestId = Activity.Current?.Id ?? HttpContext.TraceIdentifier;
29         }
30     }
31 }
```



```
1 using System;
2 using System.Collections.Generic;
3 using System.Diagnostics;
4 using System.Linq;
5 using System.Threading.Tasks;
6 using Microsoft.AspNetCore.Mvc;
7 using Microsoft.AspNetCore.Mvc.RazorPages;
8 using Microsoft.Extensions.Logging;
9
10 namespace WebStorageSample.Pages
11 {
12     [ResponseCache(Duration = 0, Location = ResponseCacheLocation.None, NoStore = true)]
13     public class ErrorModel : PageModel
14     {
15         public string RequestId { get; set; }
16     }
17 }
```

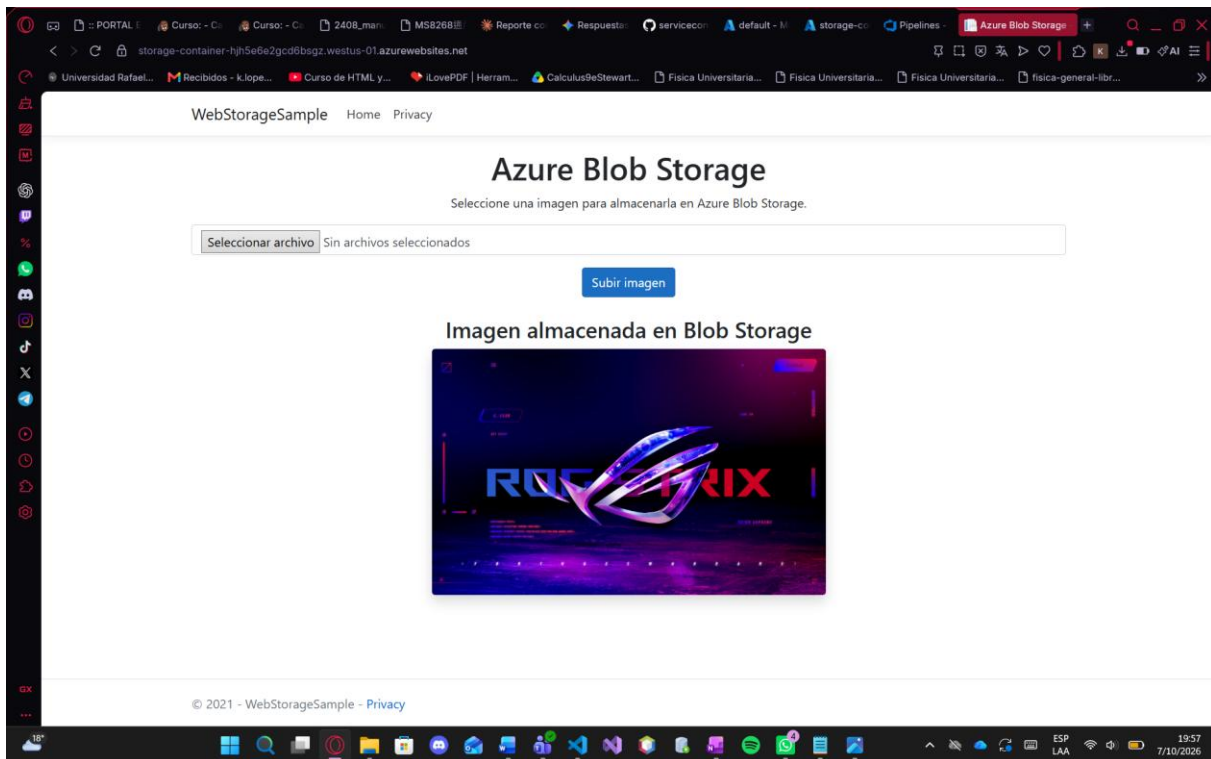
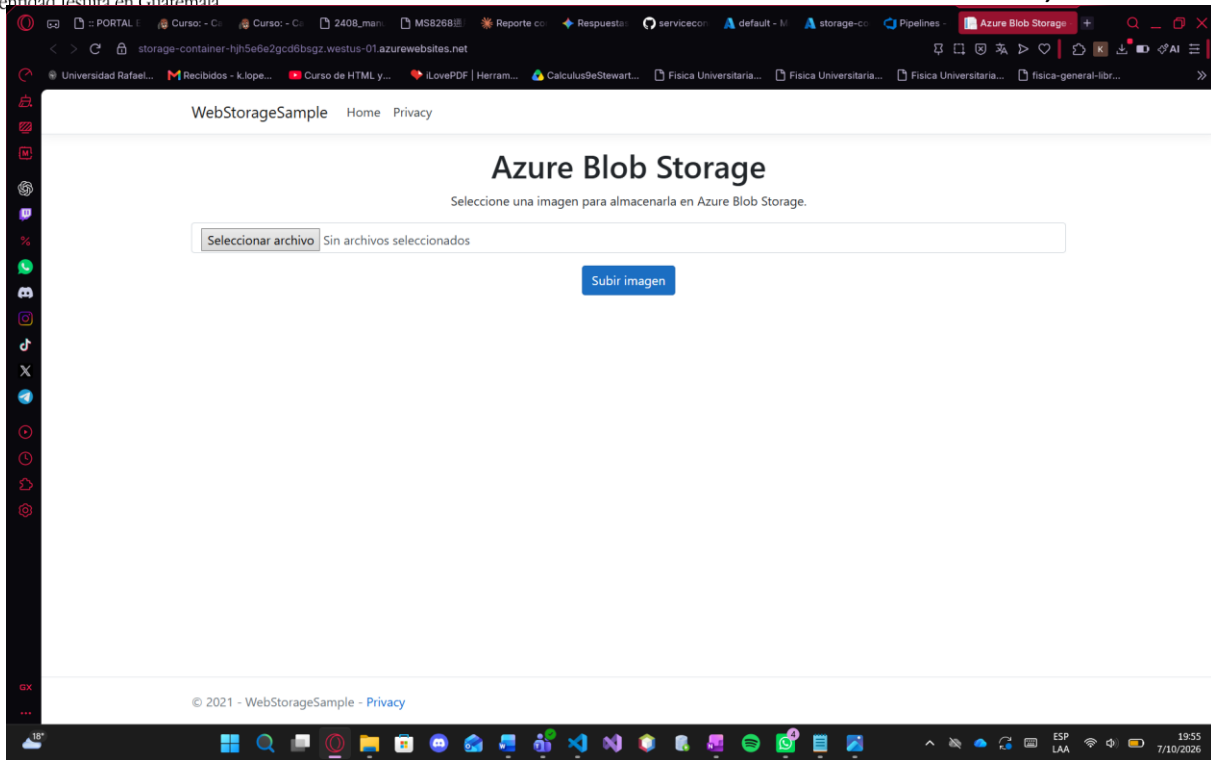
(use "git add <file>..." to update what will be committed)
(use "git restore <file>..." to discard changes in working directory)
modified: Pages/Index.cshtml
modified: Pages/Index.cshtml.cs
no changes added to commit (use "git add" and/or "git commit -a")
PS C:\Users\VASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2> git add Pages/Index.cshtml
PS C:\Users\VASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2> git add Pages/Index.cshtml.cs
PS C:\Users\VASus\url-mia-2026\semana - 14\serviceconnector-webapp-storageblob-dotnet2> git commit -m "Feat(blob): anan
iendo funcionalidad para carga de imagenes"
[main b7b7ccd] feat(blob): ananiedo funcionalidad para carga de imagenes
2 files changed, 148 insertions(+), 15 deletions(-)
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 32 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 1.63 KiB | 1.63 MiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/kilopezc/serviceconnector-webapp-storageblob-dotnet2.git

Description	Stages	Run by	Time
#20261008.3 • feat(blob): ananiedo funcionalidad para carga de imag...	Just now	Individual CI for main b7b7ccd1	47s
#20261008.2 • feat(blob): pipeline yaml change	17m ago	Individual CI for main 763788e4	2m 8s
#20261008.1 • feature: agregar pipeline	2h ago	Individual CI for main aa9af24e	3m 15s
#20261007.1 • Create azure-pipelines.yml	2h ago	Manually run by main 5910c19c	<1s



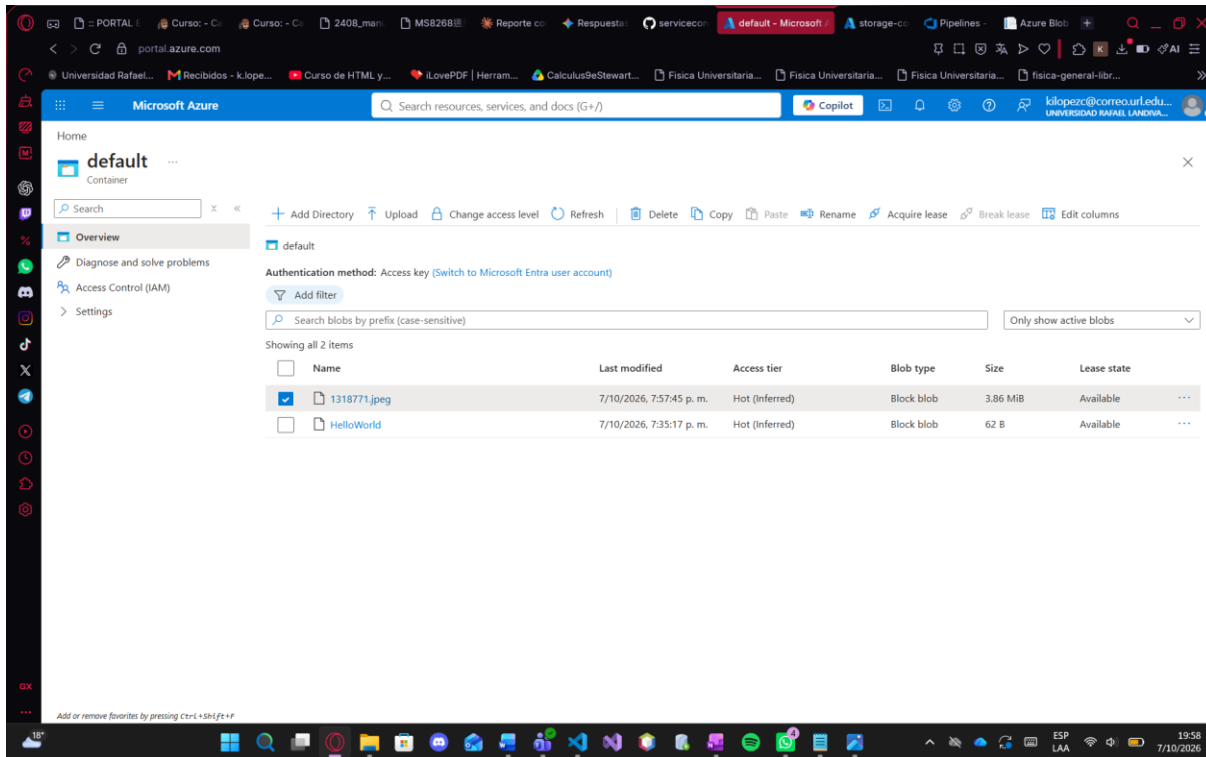
The screenshot shows the Azure DevOps web interface. The main view is the 'Runs' tab for a pipeline named 'kilopecz.serviceconnector-webapp-storageblob-dotnet2'. A modal dialog titled 'Rename/move pipeline' is open in the center. The dialog has a 'Name' field containing 'mia-blobconnector-ci-cd' and a 'Select folder' dropdown menu. The background shows a list of pipeline runs with columns for Description, Stages, and a status icon. The top navigation bar includes 'Overview', 'Boards', 'Repos', 'Pipelines', 'Environments', 'Library', 'XAML', 'Test Plans', and 'Artifacts'. The bottom status bar shows the time as 19:49 on 7/10/2026.

The screenshot shows the Azure DevOps web interface after the pipeline has been renamed. The main view is the 'Runs' tab for the pipeline named 'mia-blobconnector-ci-cd'. The background shows a list of pipeline runs with columns for Description, Stages, and a status icon. The top navigation bar includes 'Overview', 'Boards', 'Repos', 'Pipelines', 'Environments', 'Library', 'XAML', 'Test Plans', and 'Artifacts'. The bottom status bar shows the time as 19:54 on 7/10/2026.





- d) Verificar que los cambios se representen en storage account: Azure Portal → Storage Account → Storage Browser → Blob containers → Seleccionar su container → Verificar cambios



10. Preguntas de analisis

1. ¿Para qué sirve el archivo .env?

Sirve para guardar variables de configuración (como enlaces o nombres) fuera del código fuente, evitando exponer datos sensibles.

2. ¿Por qué usar Blob Storage y no guardar los archivos dentro del proyecto?

Porque la nube ofrece almacenamiento ilimitado y seguro; si la aplicación se reinicia o borra, los archivos en la nube no se pierden.

3. Diferencia entre el endpoint de Blob Storage y la URL de App Service:

El endpoint de Blob Storage es la dirección exclusiva para guardar/leer archivos, mientras que la URL de App Service es la dirección de la página web donde navega el usuario.

4. ¿Por qué usar Identidad Administrada en lugar de una Storage Account Key?

Es más segura porque elimina la necesidad de escribir contraseñas o claves en el código; Azure maneja la autenticación automáticamente.

5. Función de GitHub, Azure DevOps y Service Connection:

GitHub: Guarda y gestiona el código fuente.

Azure DevOps: Automatiza las pruebas y el despliegue de la aplicación.

Service Connection: Es el puente de permiso que autoriza a Azure DevOps a desplegar en tu cuenta de Azure.

6. ¿Qué ocurre después de hacer push a main?

Se activa automáticamente el pipeline de Azure DevOps, el cual compila el código actualizado y lo despliega en la nube.



7. Si el pipeline termina bien pero la app da error 403 al subir un archivo, ¿qué revisarías?

Los permisos de acceso (RBAC) en el Storage Account, confirmando que la identidad de la aplicación tenga el rol Storage Blob Data Contributor.

Entrega

Dentro del repositorio de GitHub, cree la carpeta semana-14/lab1/ y coloque en ella un PDF con la solución del laboratorio.

Incluya también los archivos de respaldo generados durante la práctica, así como los pantallazos que evidencien las pruebas realizadas.

Enviar link de github a plataforma de la universidad.